

PHIL 3350-001: History and Philosophy of Science
Spring Semester 2025

Course Description

PHIL 3350-001 fulfills an Upper Division Communication/Writing (CW) credit. This course—History and Philosophy of Science—covers theories about the aims and methods of science over time. That is, students will learn about the historical development of science. Using examples from the history of science, students will uncover different views of science. Topics include the relations between the sciences, between science and logic, and between science and religion.

Course Outcomes and Objectives

Investigations of the world have long circled around the same questions, such as: What is causation? What is evidence? How should science be demarcated from other forms of inquiry? That being said, the standards for what count as good answers to such questions have changed. We will begin by looking at different scientific conflicts in order to understand what has, historically, been considered as the right way to discover the truth about the natural world. To that end, from the scientific revolution, through logical positivism and Darwinian evolution, students in this class will interrogate key moments in the history of science by investigating the work of figures such as Aristotle, Galileo, and Darwin. These primary texts will be considered using methods modeled by philosophers of science and applied to cases in the history of science. We will investigate scientific paradigm shifts, what constitutes a scientific theory, scientific disagreements and advances, and pitfalls. Students will also learn basic deductive logic and contemporary philosophy of well-being.

Course Requirements

- 12.5%: Prospective Idea 1
- 12.5%: Prospective Idea 2
- 20%: Module Quizzes
- 25%: Weekly Discussion Posts
- 30%: Term Paper / Final Exam

Attendance and Participation

Show up on time and ready to engage. Students are allowed one free, unjustified, absence. Each absence after that will cost 1.5 percentage points from the final grade.

Module Quizzes

Each module will be about three weeks long. At the end of each module, there will be a 7-10 question quiz (short answer and multiple choice) regarding the information in the module. There will be no retakes. The quiz with the lowest grade will be omitted.

Weekly Discussion Posts

Along with a summary of the material, discussion will be based on the discussion questions you post each week. Questions will be graded on a scale of 1 (more effort needed) to 2 (fine job) to 3 (great work) to 4 (truly impressive). Questions covering matters of fact (such as the clarification of a concept, the identity of a historical player, etc.) will in most cases be graded lower than those addressing matters of interpretation (such as a criticism of an argument, an inquiry into the relation of material covered to other course material, etc.). Expected length: 100 - 150 words. These are due by 9pm on the Monday before our Tuesday class at the beginning of each week. There will be 13 weekly assignments, starting on Week 2 and ending on Week 14. Each student may choose to ignore one weekly assignment for free. Each one missing or incomplete after that will cost 1.5 percentage points of the Weekly Assignments grade.

Prospective Ideas

Students will be asked to undergo the beginning stages of writing a paper—twice. When submitting a “prospective idea” in its gradable form, it will not be a full paper, but rather the idea of a full paper in the form of a developed outline or rough draft. The first prospective idea will mainly draw on one’s intuitive interests as one enters the class; it will be due September 29th. The second prospective idea will draw more majorly on the material we cover in class, due on November 10th. It may be inspired by one’s first prospective idea, but students are encouraged to conjure new ideas as well.

Term Paper

The term paper is the main product of this class. The final paper must draw on one of the two prospective ideas, and be a maximum of 3500 words. While the students are expected to present their term papers, they will not be graded on this presentation further than the “participation” grade. The term paper will be due on Sunday, December 8th.

Students will have the option of taking a final exam, rather than submitting a final paper. The exam will consist of written responses to concept identification questions and essay prompts. If students fail to submit at least one prospective idea, they default to taking the final exam.

Grading Scale

A+ = 100-97	C+ = 79-77
A = 96-93	C = 76-73
A- = 92-90	C- = 72-70
B+ = 89-87	D+ = 69-67
B = 86-83	D = 66-63
B- = 82-80	D- = 62-60
	F = 59-0

University Policies

Americans With Disabilities Act (ADA)

The University of Utah seeks to provide equal access to its programs, services, and activities for people with disabilities.

All written information in this course can be made available in an alternative format with prior notification to the Center for Disability & Access (CDA). CDA will work with you and the instructor to make arrangements for accommodations. Prior notice is appreciated. To read the full accommodations policy for the University of Utah, please see Section Q of the Instruction & Evaluation regulations.

In compliance with ADA requirements, some students may need to record course content. Any recordings of course content are for personal use only, should not be shared, and should never be made publicly available. In addition, recordings must be destroyed at the conclusion of the course.

If you will need accommodations in this class, or for more information about what support they provide, contact:

Center for Disability & Access

801-581-5020

disability.utah.edu

65 Student Services Building

201 S 1460 E

Salt Lake City, UT 84112

Safety at the U

The University of Utah values the safety of all campus community members. You will receive important emergency alerts and safety messages regarding campus safety via text message. For more safety information and to view available training resources, including helpful videos, visit safeu.utah.edu.

To report suspicious activity or to request a courtesy escort, contact:

Campus Police & Department of Public Safety

801-585-COPS (801-585-2677)

dps.utah.edu

1735 E. S. Campus Dr.

Salt Lake City, UT 84112

Addressing Sexual Misconduct

Title IX makes it clear that violence and harassment based on sex and gender (which includes sexual orientation and gender identity/expression) is a civil rights offense subject to the same kinds of accountability and the same kinds of support applied to offenses against other protected categories such as race, national origin, color, religion, age, status as a person with a disability, veteran's status, or genetic information.

If you or someone you know has been harassed or assaulted, you are encouraged to report it to university officials:

Title IX Coordinator & Office of Equal Opportunity and Affirmative Action

801-581-8365

oeo.utah.edu

135 Park Building

201 Presidents' Cir.

Salt Lake City, UT 84112

Office of the Dean of Students

801-581-7066

deanofstudents.utah.edu

270 Union Building

200 S. Central Campus Dr.
Salt Lake City, UT 84112

To file a police report, contact:

Campus Police & Department of Public Safety

801-585-COPS (801-585-2677)

dps.utah.edu

1735 E. S. Campus Dr.
Salt Lake City, UT 84112

If you do not feel comfortable reporting to authorities, the U's Victim-Survivor Advocates provide free, confidential, and trauma-informed support services to students, faculty, and staff who have experienced interpersonal violence.

To privately explore options and resources available to you with an advocate, contact:

Center for Campus Wellness

801-581-7776

wellness.utah.edu

350 Student Services Building
201 S. 1460 E.
Salt Lake City, UT 84112

Academic Misconduct

It is expected that students comply with University of Utah policies regarding academic honesty, including but not limited to refraining from cheating, plagiarizing, misrepresenting one's work, and/or inappropriately collaborating. This includes the use of generative artificial intelligence (AI) tools without citation, documentation, or authorization. Students are expected to adhere to the prescribed professional and ethical standards of the profession/discipline for which they are preparing. Any student who engages in academic dishonesty or who violates the professional and ethical standards for their profession/discipline may be subject to academic sanctions as per the University of Utah's Student Code: Policy 6-410: Student Academic Performance, Academic Conduct, and Professional and Ethical Conduct.

Plagiarism and cheating are serious offenses and may be punished by failure on an individual assignment, and/or failure in the course. Academic misconduct, according to the University of Utah Student Code:

“...Includes, but is not limited to, cheating, misrepresenting one’s work, inappropriately collaborating, plagiarism, and fabrication or falsification of information...It also includes facilitating academic misconduct by intentionally helping or attempting to help another to commit an act of academic misconduct.”

For details on plagiarism and other important course conduct issues, see the U's Code of Student Rights and Responsibilities.

Preliminary Course Schedule

Module 1: Some History of Science: What is it, and where does it come from?

January 6th - January 24th

Week 1: Introduction

- *January 6th*
 - Read:
 - Syllabus
- *January 8th and 10th*
 - Read
 - Godfrey-Smith, P. (2003), “Introduction” (1.1-1.4) from *Theory and Reality: An Introduction to the Philosophy of Science*
 - Aristotle's Four Causes Passage

Week 2: Galileo and the Scientific Revolution

- *January 13th - 17th*
 - Read:
 - “Galileo” from Matthews, M. (1989) from *The Scientific Background to Modern Philosophy*
 - Check out these videos!
 - https://www.youtube.com/watch?v=frZ9dN_ATew
 - <https://www.youtube.com/watch?v=9BQKe9HT1OE>

Review: Prospectus 1 Assignment

Week 3: Darwin and a Revolution in Biology

- *January 20th*
 - HOLIDAY: MLK Jr. Day
- *January 22nd*
 - Read:
 - Sober, E. (2000), Chapter 2: “Creationism” in *Philosophy of Biology* (pp. 27-57)
- *January 24th*
 - Read: Excerpt from Darwin, “Tangled Bank”

Module 2: 20th Century Philosophy of Science

January 27 - February 10th

Week 4: Logical Positivism

- *January 27th*
 - Read:
 - Hempel, C. (1966) “Laws and their Role in Scientific Explanation” (pp. 47-69)
- *January 29th*
 - Cartwright, N (1980) “The Truth Doesn’t Explain Much” (pp. 159-163).
- *January 31st*
 - No in-person class.
 - Optional Reading:
 - Downes, S. (1997), “Logical Positivism/Logical Empiricism” (pp. 218-224).
 - **Write on the discussion board: one post plus one response.**

Week 5: Falsification

- *February 3rd - 7th*
 - Read:
 - Popper, K. (1963) Chapter 1: “Science: Conjectures and Refutations” (pp. 1-49)

Week 6: Workshop

- *February 10th*
 - Paper-writing workshop
- *February 12 - 14th*
 - TBD

Prospectus 1 is due by 11:59pm, Sunday, February 16th.

Module 3: Script Flips

February 17th - March 7th

Week 7: What is the actual problem with induction?

- *February 17th*
 - HOLIDAY: President's Day
- *February 19th*
 - Read:
 - Okasha 2001: What Did Hume Really Show About Induction?
- *February 21st*
 - No in-person class.
 - Start reading:
 - Kuhn 1962: Structure of Scientific Revolutions—Ch. 9
 - **Write on the discussion board: one post plus one response.**

Week 8: Paradigm Shifts

- *February 24th*
 - Read:
 - Kuhn 1962: Structure of Scientific Revolutions—Ch. 9
- *February 26th*
 - Read:
 - Lakatos: Falsification and the Methodology of Scientific Research Programmes
- *February 28th*
 - ...continued

Week 9: Human Nature?

- *March 3rd*
 - Read:
 - Hull 1986: On Human Nature
- *March 5th*
 - Read
 - Washington 2016: Culturally Unbound
- *March 7th*
 - Read
 - Amundson (2000): Against Normal Function

Week 10: SPRING BREAK

March 9 -16th

Module 4: Philosophy of Self-Control

March 17 - April 4th

Week 11: Internal Values

- *March 17th*
 - Read:
 - *Berkman et al 2017, Self-Control as Value-Based Choice*
- *March 19th*
 - Read:
 - *Washington 2018, Contextualism as a Solution to Paternalism in Psychiatric Practice*
- *March 21st*
 - Optional Reading:
 - *Kuhn 1962: Structure of Scientific Revolutions—Ch. 10*
 - Continue from previous two classes

Week 12: Internal Unreliability

- *March 24st*
 - Read:
 - Read: *Saunders et al. 2018, Reported Self-control is not Meaningfully Associated with Inhibition-related Executive Function: A Bayesian Analysis; feel free to ignore the methods section*
- *March 26th*
 - Read:
 - Read: *Schwitzgebel 2008, The Unreliability of Naive Introspection; read at least 10 pages*
- *March 28th*
 - Read:
 - *Robinson and Adinoff 2016, The Classification of Substance Use Disorders: Historical, Contextual, and Conceptual Considerations. Pages 8 - 18*

Week 13: Modern Views of Addiction

- *March 31st*
 - Read:

- *Sripada 2021*, Impaired control in addiction involves cognitive distortions and unreliable self-control, not compulsive desires and overwhelmed self-control
- *April 2nd*
 - Read:
 - *Leshner 1997*, Addiction is a Brain Disease, and It Matters
 - *Pickard 2021*, Is addiction a brain disease? A plea for agnosticism and heterogeneity
- *April 4th*
 - No in-person class.
 - Write on the discussion board. One post plus one response.

Prospectus 2 is due by 11:59pm, Sunday, April 6th.

Last Two Weeks

November April 7th - April 21

- Presenting Papers

Term Paper is due by 11:59pm, Sunday, April 27th.

Grades Due: Monday May 12th